Component-wise Approximate Bayesian Computation via Gibbs-like steps

Comparative study of ABC-Gibbs, SMC-ABC and Adaptive Random-Walk Metropolis on hierarchical models

Project 4 — *Gibbs and ABC*

Simulation and Monte Carlo Course
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Abstract

This project provides a rigorous comparative study of three Monte Carlo algorithms on two hierarchical models, in accordance with the three-algorithm structure of the prompt. The first algorithm is the ABC-Gibbs sampler introduced by Clarté *et al.* (2021); the second is an adaptive SMC-ABC sampler that follows Algorithm 5 of the paper's supplement (Del Moral *et al.*, 2012; Toni *et al.*, 2008), enhanced with a Correlated Pseudo-Marginal scheme (Deligiannidis *et al.*, 2018) and a robust Cholesky regularisation for the mutation kernel; the third is an Adaptive Random-Walk Metropolis sampler (Haario *et al.*, 2001) which exploits the closed-form Gaussian log-likelihood of the Normal-Normal model and serves as a *dynamic* ground truth alongside the analytical conjugate posterior. The two ground truths cross-validate each other and provide a noise-free benchmark for evaluating ABC error. The comparison is based on three metric families: (i) computational cost, (ii) inferential error against the analytical posterior, (iii) Monte Carlo error via batch means and inter-run replicates. On the hierarchical Gaussian model in dimension $1 + n = 21$, ABC-Gibbs vastly outperforms SMC-ABC at half the simulation cost: average Wasserstein-1 distance on the μ_j's of 0.024 versus 0.922. On the simple hierarchical g-and-k model in dimension $1 + n = 51$, the predictive distance is 352 versus 1082, confirming the progressive collapse of SMC-ABC in increasing dimension. These results quantitatively reproduce the findings of Clarté *et al.* (2021) and illustrate how the component-wise strategy breaks the curse of dimensionality in likelihood-free inference.

Contents

1 Introduction and Problem Statement

1.1 General Framework of Likelihood-Free Inference

Classical Bayesian inference relies on the evaluation of the posterior

$$\pi(\theta \mid x^\star) \;\propto\; \pi(\theta)\, f(x^\star \mid \theta),$$

where π denotes the prior on $\theta \in \Theta$ and $f(\cdot \mid \theta)$ the likelihood. In many modern models — generative models in population genetics (Tavaré *et al.*, 1997; Beaumont *et al.*, 2002), biological dynamical systems (Toni *et al.*, 2008), g-and-k distributions (Prangle, 2017) — the likelihood cannot be evaluated in reasonable time, but a *simulator* is available: given θ, one can produce $x \sim f(\cdot \mid \theta)$. This is the setting of LIKELIHOOD-FREE or APPROXIMATE BAYESIAN COMPUTATION (ABC).

The core idea, pioneered by Tavaré *et al.* (1997) and formalised by Beaumont *et al.* (2002), is to draw pairs $(\theta^{(i)}, x^{(i)})$ from the prior predictive $\pi(\theta)f(x \mid \theta)$ and retain those for which $x^{(i)}$ falls in a neighbourhood of $x^\star$, typically through a summary statistic $s(\cdot)$ and a tolerance $\varepsilon > 0$.

1.2 The Curse of Dimensionality in ABC

The major drawback of vanilla ABC is its dramatic sensitivity to parameter dimension. Fearnhead & Prangle (2012) and Li & Fearnhead (2018) show that for an optimal approximation, the dimension of s must coincide with that of θ. Yet, at fixed summary dimension, the probability that a simulation falls within a Euclidean ε-ball around $s(x^\star)$ decreases exponentially with that dimension, leading to a prohibitive computational cost.

Clarté *et al.* (2021) proposed a strategy to break this curse: exploit the conditional structure of the joint posterior via Gibbs-like steps, each updating a single component θ_j on a low-dimensional summary s_j. This approach, dubbed ABC-GIBBS, sacrifices some classical Gibbs guarantees (the conditionals may be incompatible) in favour of substantial numerical efficiency.

1.3 Purpose of the Project

In accordance with the instructions of *Project 4 — Gibbs and ABC*, we compare *three* algorithms on the hierarchical Gaussian model:

1. the ABC-GIBBS algorithm by Clarté *et al.* (2021) (reference algorithm);
2. an adaptive SMC-ABC sampler (Algorithm 5 of the supplement of Clarté *et al.* (2021), fusing Del Moral *et al.* (2012) for the adaptive ESS schedule and Toni *et al.* (2008) for the Gaussian mutation kernel), enhanced with a Correlated Pseudo-Marginal scheme (Deligiannidis *et al.*, 2018) and a robust Cholesky regularisation. SMC-ABC is the principal competitor used in the reference paper;
3. an **Adaptive Random-Walk Metropolis** sampler (Haario *et al.*, 2001) on the joint state $\theta = (\alpha, \mu_1, \ldots, \mu_n)$. It exploits the closed-form Gaussian log-likelihood of the model and acts as a *dynamic* ground truth (ν_0 in the language of Theorem 3 of the paper).

The Normal-Normal model further admits an analytical *conjugate* posterior (Proposition 3.1), which we use as a *noise-free static* ground truth. The two references — one MCMC-based, one closed-form — cross-validate each other; their empirical agreement, documented in Section 6.1, makes this dual benchmark significantly more rigorous than either alone. For the g-and-k model, no analytical posterior is available and the likelihood is intractable, so we fall back on the predictive metric of Clarté *et al.* (2021).

Three metric families are systematically reported, as the prompt requires: (i) CPU time and number of model simulations; (ii) inferential error (bias, variance ratio, Wasserstein-1 distance); (iii) intra-chain Monte Carlo error via batch means (Flegal & Jones, 2010). An independent replicate study confirms the relevance of these MC-error estimates. Finally, as a bonus, we extend the comparison to the simple hierarchical g-and-k model from Section 4 of the paper, illustrating the robustness of the conclusion on a truly intractable model.

2 Theoretical Framework: ABC, Gibbs, ABC-Gibbs

2.1 Vanilla ABC

Definition 2.1. ABC Distribution

Given a summary statistic $s : \mathcal{X} \to \mathbb{R}^q$, a distance d on $\mathbb{R}^q$, and a tolerance $\varepsilon > 0$, the *ABC distribution* is the distribution on Θ with density

$$\pi_\varepsilon(\theta \mid s(x^\star)) \;\propto\; \pi(\theta) \int_{\mathcal{X}} f(x \mid \theta) \mathbb{1}\big\{d\big(s(x), s(x^\star)\big) < \varepsilon\big\} \,\mathrm{d}x.$$

Proposition 2.2. Limit of ABC methods

Under standard regularity assumptions (Frazier *et al.*, 2018), as $\varepsilon \downarrow 0$, $\pi_\varepsilon(\cdot \mid s(x^\star))$ converges in total variation to $\pi(\cdot \mid s(x^\star))$, which coincides with the exact posterior $\pi(\cdot \mid x^\star)$ if and only if s is sufficient for θ.

Algorithm 1: Rejection ABC (vanilla)

Input: observations $x^\star$, size N, tolerance $\varepsilon > 0$, statistic s, distance d.
Output: sample $(\theta^{(1)}, \dots, \theta^{(N)})$.

```
for i = 1, ..., N do
    repeat
        Draw θ^(i) ~ π(·);
        Draw x^(i) ~ f(· | θ^(i));
    until d(s(x^(i)), s(x*)) < ε;
end
```

2.2 Gibbs Sampler

Definition 2.3. Gibbs Sampler

Let $\theta = (\theta_1, \dots, \theta_n)$ and π be a target distribution on $\Theta = \prod_{j=1}^n \Theta_j$ admitting simulatable full conditionals $\pi(\theta_j \mid \theta_{-j})$. The *systematic Gibbs sampler* produces a Markov chain $(\theta^{(t)})_{t \geq 0}$ via cyclical updates:

$$\theta_j^{(t)} \sim \pi\Big(\cdot \mid \theta_1^{(t)}, \dots, \theta_{j-1}^{(t)}, \theta_{j+1}^{(t-1)}, \dots, \theta_n^{(t-1)}\Big), \qquad j = 1, \dots, n.$$

The major appeal of the Gibbs sampler is its match with hierarchical structures. Under the model

$$x_j \mid \mu_j \sim f(\cdot \mid \mu_j), \qquad \mu_j \mid \alpha \overset{\text{iid}}{\sim} \pi(\cdot \mid \alpha), \qquad \alpha \sim \pi(\alpha), \tag{H}$$

the conditional posterior of $\mu = (\mu_1, \dots, \mu_n)$ given α factorises as $\prod_j \pi(\mu_j \mid \alpha) f(x_j \mid \mu_j)$, decomposing into n independent 1-dimensional problems. When f is intractable the classical Gibbs sampler fails, and ABC-Gibbs steps in.

2.3 ABC-Gibbs

Algorithm 2: ABC-Gibbs (Clarté *et al.*, 2021)

Input: $x^\star$, size N, initial point $\theta^{(0)}$, thresholds $\varepsilon_1, \dots, \varepsilon_n$, summaries $s_1, \dots, s_n$, distances $d_1, \dots, d_n$.

Output: $(\theta^{(1)}, \dots, \theta^{(N)})$.

```
for i = 1, ..., N do
    for j = 1, ..., n do
        Draw θ_j^(i) ~ π_{ε_j}{· | s_j(x*, θ_1^(i), ..., θ_{j-1}^(i), θ_{j+1}^(i-1), ..., θ_n^(i-1))};
    end
end
```

$$\text{Draw } \theta_j^{(i)} \sim \pi_{\varepsilon_j}\left\{\cdot \mid s_j\left(x^\star, \theta_1^{(i)}, \dots, \theta_{j-1}^{(i)}, \theta_{j+1}^{(i-1)}, \dots, \theta_n^{(i-1)}\right)\right\};$$

The algorithm replaces each call to the exact full conditional with its ABC counterpart: a mini-reference table is drawn from the *conditional* predictive (rather than the prior predictive), and the candidate minimising the summary distance is retained. Since each θ_j is typically 1-dimensional, s_j can be too (Fearnhead & Prangle, 2012), and the conditional predictive is centred on the current values of the other components, which makes the procedure dramatically more efficient than vanilla ABC in hierarchical settings.

2.4 Convergence Theorem (case $n = 2$)

Theorem 2.4. Convergence of ABC-Gibbs, $n = 2$ (Th. 1, Clarté *et al.*)

Suppose there exists $0 < \kappa < 1/2$ such that

$$\sup_{\theta_1, \tilde{\theta}_1} \left\| \pi_{\varepsilon_2}\{\cdot \mid s_2(x^\star, \theta_1)\} \; - \; \pi_{\varepsilon_2}\left\{\cdot \mid s_2(x^\star, \tilde{\theta}_1)\right\} \right\|_{\mathrm{TV}} \; = \; \kappa.$$

Then the Markov chain produced by Algorithm 2 converges geometrically in total variation to a stationary distribution ν_ε, with rate $1 - 2\kappa$.

Proof. We outline the proof, omitting the conditioning on $x^\star$, s_1, s_2 to lighten the notation. Let Q be the transition kernel of the marginal chain on Θ_1. We show that Q is a contraction with constant $1 - 2\kappa$ in total variation; Banach's fixed-point theorem yields existence and uniqueness of ν_ε together with geometric convergence.

Construction of a coupling. For two measures $\nu, \tilde{\nu} \in \mathcal{P}(\Theta_1)$, let ξ_0 be the optimal coupling between ν and $\tilde{\nu}$, with $\mathbb{P}_{\xi_0}(\theta_1 \neq \tilde{\theta}_1) = 2\|\nu - \tilde{\nu}\|_{\mathrm{TV}}$ (existence on Polish spaces). Define $\widetilde{Q}$ by (i) $(\theta_1, \tilde{\theta}_1) \sim \xi_0$; (ii) $(\theta_2, \tilde{\theta}_2) \sim \xi_1(\cdot \mid \theta_1, \tilde{\theta}_1)$ optimal coupling of $\pi_{\varepsilon_2}(\cdot \mid \theta_1)$ and $\pi_{\varepsilon_2}(\cdot \mid \tilde{\theta}_1)$; (iii) $(\theta_1', \tilde{\theta}_1') \sim \xi_2(\cdot \mid \theta_2, \tilde{\theta}_2)$ optimal coupling of $\pi_{\varepsilon_1}(\cdot \mid \theta_2)$ and $\pi_{\varepsilon_1}(\cdot \mid \tilde{\theta}_2)$. The construction satisfies the absorption property: if $\theta_2 = \tilde{\theta}_2$, then $\theta_1' = \tilde{\theta}_1'$ a.s.

Decoupling probability. By the coupling characterisation of total variation:

$$\|Q\nu - Q\tilde{\nu}\|_{\mathrm{TV}} \leq \tfrac{1}{2}\mathbb{P}_{\widetilde{Q}\xi_0}(\theta_1' \neq \tilde{\theta}_1').$$

Decomposing on $\{\theta_1 = \tilde{\theta}_1\}$ and using absorption,

$$\|Q\nu - Q\tilde{\nu}\|_{\mathrm{TV}} \leq \|\nu - \tilde{\nu}\|_{\mathrm{TV}}\, \mathbb{P}(\theta_1' \neq \tilde{\theta}_1' \mid \theta_1 \neq \tilde{\theta}_1).$$

By absorption, decoupling at step 3 is bounded by decoupling at step 2: $\mathbb{P}(\theta_2 \neq \tilde{\theta}_2 \mid \theta_1, \tilde{\theta}_1) = 2\|\pi_{\varepsilon_2}(\cdot \mid \theta_1) - \pi_{\varepsilon_2}(\cdot \mid \tilde{\theta}_1)\|_{\mathrm{TV}} \leq 2\kappa$. Integrating yields $\mathbb{P}(\theta_2 \neq \tilde{\theta}_2 \mid \theta_1 \neq \tilde{\theta}_1) \leq 2\kappa$.

Conclusion. $\|Q\nu - Q\tilde{\nu}\|_{\mathrm{TV}} \leq (1 - 2\kappa)\|\nu - \tilde{\nu}\|_{\mathrm{TV}}$. Since $\kappa < 1/2$, Q is a strict contraction on the complete space $(\mathcal{P}(\Theta_1), \|\cdot\|_{\mathrm{TV}})$. Banach's theorem yields the unique fixed point $\nu_\varepsilon^{(1)}$. □

2.5 Extension to $n > 2$ and limit bound

Theorem 2.5. Convergence in the general case (Th. 2)

Suppose that for all $\ell \leq n$,

$$\kappa_\ell = \sup_{\theta_{>\ell}, \tilde{\theta}_{>\ell}} \sup_{\theta_{<\ell}} \left\| \pi_{\varepsilon_\ell}\{\cdot \mid s_\ell(x^\star, \theta_{<\ell}, \theta_{>\ell})\} - \pi_{\varepsilon_\ell}\left\{\cdot \mid s_\ell(x^\star, \theta_{<\ell}, \tilde{\theta}_{>\ell})\right\} \right\|_{\mathrm{TV}} < \tfrac{1}{2}.$$

Then the chain of Algorithm 2 converges geometrically to ν_ε at rate $1 - \prod_\ell 2\kappa_\ell$.

Theorem 2.6. Distance between the limit distribution and the ABC posterior (Th. 3)

For $n = 2$, set

$$L_0 = \sup_{\varepsilon_2, \theta_1, \tilde{\theta}_1} \|\pi_{\varepsilon_2}(\cdot \mid s_2(x^\star, \theta_1)) - \pi_0(\cdot \mid s_2(x^\star, \tilde{\theta}_1))\|_{\mathrm{TV}} < \tfrac{1}{2},$$

$$L_j(\varepsilon_j) \xrightarrow[\varepsilon_j \to 0]{} 0 \quad \text{for } j = 1, 2.$$

Then $\|\nu_\varepsilon - \nu_0\|_{\mathrm{TV}} \leq \frac{L_1(\varepsilon_1) + L_2(\varepsilon_2)}{1 - 2L_0} \xrightarrow[\varepsilon \to 0]{} 0$.

2.6 Hierarchical Case: Theorem 4

For model (H) with a single layer where $\pi(\mu \mid x^\star, \alpha)$ is exactly simulatable (which holds for the Normal-Normal case), a stronger result is available.

Theorem 2.7. Hierarchical case with exact conditional (Th. 4)

Suppose there is a non-empty convex set C with positive prior measure such that $\kappa_1, \kappa_2, \kappa_3 > 0$ where these constants are the (uniform) lower bounds on $\pi(B_{s_\alpha(\mu), \varepsilon_\alpha/4})$, on $\pi_{\varepsilon_\mu}\{B_{s_\alpha(\mu), 3\varepsilon_\alpha/2} \mid s_\mu(x^\star, \alpha), \alpha\}$, and on $\pi_{\varepsilon_\mu}\{s_\alpha(\mu) \in C \mid s_\mu(x^\star, \alpha)\}$ respectively. Then the chain converges geometrically to ν_ε at rate $1 - \kappa_1\kappa_2\kappa_3^2$.

2.7 Pathology: an illuminating counter-example

The paper presents an instructive counter-example (supplement, Section 12): a single observation $x^\star = 5$ from a mixture $\frac{1}{2}\mathcal{U}(\theta_1, \theta_1 + 1) + \frac{1}{2}\mathcal{U}(\theta_2, \theta_2 + 1)$, with uniform prior on $A = \{(\theta_1, \theta_2) : 0 \leq \theta_1, \theta_2 \leq 10, |\theta_1 - \theta_2| > 2\}$. The exact posterior is uniform over the union of two symmetric bands; for $\varepsilon = 0.5$, vanilla ABC captures both, but ABC-Gibbs only captures one (depending on initialisation), since the supports of the two conditionals can be disjoint, violating $\kappa < 1/2$. The component-wise strategy is therefore not a universal cure: it breaks down on structured multimodalities.

3 Studied Models

3.1 Model 1: Hierarchical Gaussian (Section 3.2)

The first model strictly follows Section 3.2 of the paper:

$$\mu_j \overset{\text{iid}}{\sim} \mathcal{N}(\alpha, \varsigma^2), \qquad x_{j,k} \mid \mu_j \overset{\text{ind}}{\sim} \mathcal{N}(\mu_j, \sigma^2), \qquad j = 1, \ldots, n, \quad k = 1, \ldots, K, \tag{N}$$

with hyperprior $\alpha \sim \mathcal{U}[-4, 4]$ and known variances. We match Figure 2 of the paper exactly: $\sigma = \varsigma = 1$, $K = 10$, $n = 20$, dimension $1 + n = 21$.

Summary statistics. We use empirical means, which are sufficient: $s_\mu(x_j^\star) = \bar{x}_j^\star$ for the μ_j's and $s_\alpha(\mu) = \bar{\mu}$ for α.

Proposition 3.1. Exact posterior and full conditionals

Under model (N),

$$\mu_j \mid \alpha, x^\star \sim \mathcal{N}\left(\frac{\alpha/\varsigma^2 + K\bar{x}_j^\star/\sigma^2}{1/\varsigma^2 + K/\sigma^2},\ \frac{1}{1/\varsigma^2 + K/\sigma^2}\right), \tag{1}$$

$$\alpha \mid \mu \sim \mathcal{N}\big(\bar{\mu}, \varsigma^2/n\big) \text{ truncated to } [-4, 4]. \tag{2}$$

Marginalising the μ_j's yields

$$\alpha \mid x^\star \sim \mathcal{N}\left(\frac{1}{n}\sum_{j=1}^{n} \bar{x}_j^\star,\ \frac{1}{n}\big(\varsigma^2 + \sigma^2/K\big)\right) \text{ truncated to } [-4, 4]. \tag{3}$$

Proof. The joint density is, up to a constant,

$$\pi(\alpha, \mu, x^\star) \propto \mathbb{1}_{[-4,4]}(\alpha) \prod_{j=1}^{n} \exp\left\{-\frac{(\mu_j - \alpha)^2}{2\varsigma^2}\right\} \prod_{k=1}^{K} \exp\left\{-\frac{(x_{j,k}^\star - \mu_j)^2}{2\sigma^2}\right\}.$$

Keeping only the terms in μ_j and completing the square yields (1); the same operation on α gives (2). For the marginal, conditional on α the $\bar{x}_j$'s are i.i.d. $\mathcal{N}(\alpha, \varsigma^2 + \sigma^2/K)$, hence $\bar{\bar{x}} \mid \alpha \sim \mathcal{N}(\alpha, (\varsigma^2 + \sigma^2/K)/n)$; conjugacy with the (improper) flat prior gives (3). $\square$

This analytical posterior (3)–(1) acts as a *static* ground truth in our comparisons. Section 4.3 explains how it interacts with the RWM-AM dynamic ground truth.

3.2 Model 2: Simple hierarchical g-and-k (bonus)

The second model, treated as a bonus, is the simple hierarchical g-and-k model from Section 4 of the paper:

$$\mu_i \sim \mathcal{N}(\alpha, 1), \qquad x_i \sim gk(\mu_i, B, g, k), \qquad i = 1, \ldots, n, \tag{GK}$$

with $\alpha \sim \mathcal{U}[-10, 10]$ (paper p.7), and the g-and-k distribution defined by its quantile function:

$$F^{-1}(u; \mu, B, g, k, c) = \mu + B\left(1 + c\,\frac{1 - e^{-g\,z(u)}}{1 + e^{-g\,z(u)}}\right)\big(1 + z(u)^2\big)^k z(u),$$

with $z = \Phi^{-1}$ and $c = 0.8$. The density has no closed form, but simulation is immediate by inversion (and we route the sampler directly through a Gaussian auxiliary $z \sim \mathcal{N}(0,1)$, which is essential for the Correlated Pseudo-Marginal scheme of SMC-ABC). We fix $B = 1$, $g = 0.2$, $k = 0.5$, $\alpha^\star = 3$, $n = 50$ groups, $K = 100$ observations per group.

Summary statistic. Without sufficiency, we follow Prangle (2017) and Clarté *et al.* (2021) Section 4 and use *octiles* on the 9 quantiles $\{0, 1/8, \ldots, 1\}$ with MAE distance:

$$s_\mu(x_i) = \big(q(x_i, p)\big)_{p \in \{0, 1/8, \ldots, 1\}} \in \mathbb{R}^9, \qquad d\big(s_\mu(x_i), s_\mu(x_i^\star)\big) = \sum_p \big|q(x_i, p) - q(x_i^\star, p)\big|.$$

For α we keep $s_\alpha(\mu) = \bar{\mu}$, since $\mu_j \mid \alpha$ remains Gaussian.

4 Compared Methods and Justification of Choices

4.1 ABC-Gibbs: Implementation for the Gaussian model

Under (N), Algorithm 2 unfolds as follows at each iteration $i \geq 1$:

1. For each $j = 1, \ldots, n$: simulate N_μ proposals $\mu_j^{(\ell)} \sim \mathcal{N}(\alpha^{(i-1)}, \varsigma^2)$; for each, simulate K pseudo-observations $x_{j,k}^{(\ell)} \sim \mathcal{N}(\mu_j^{(\ell)}, \sigma^2)$ and compute $\bar{x}_j^{(\ell)}$; set $\mu_j^{(i)} \leftarrow \mu_j^{(\ell^\star)}$ with $\ell^\star = \arg\min_\ell |\bar{x}_j^{(\ell)} - \bar{x}_j^\star|$.
2. For α: simulate N_α proposals $\alpha^{(\ell)} \sim \mathcal{U}[-4, 4]$; for each, simulate $\mu_j^{(\ell)} \sim \mathcal{N}(\alpha^{(\ell)}, \varsigma^2)$ and compute $\bar{\mu}^{(\ell)}$; set $\alpha^{(i)} \leftarrow \alpha^{(\ell^\star)}$ with $\ell^\star = \arg\min_\ell |\bar{\mu}^{(\ell)} - \bar{\mu}^{(i)}|$.

Strict likelihood-freeness. Although the empirical mean $\bar{x}_j^{(\ell)}$ of K Gaussian samples has the analytical distribution $\mathcal{N}(\mu_j^{(\ell)}, \sigma^2/K)$, our implementation *actually simulates* the K pseudo-observations and computes their mean. This preserves the methodological agnosticism of ABC and is essential for generalisation to models like g-and-k where no analytical statistic is available.

Calibration. As recommended in the paper, $N_\mu = N_\alpha = 30$ and $N = 1000$ Gibbs iterations (with a burn-in of 5 discarded, as in the paper p.7) provide a reasonable trade-off. The total simulation cost is

$$N\big(N_\mu \cdot n \cdot K + N_\alpha \cdot n + N_\alpha \cdot 1\big) \approx N \cdot N_\mu \cdot n \cdot K \approx 6.6 \cdot 10^6,$$

in agreement with the measured $7.23 \cdot 10^6$.

4.2 SMC-ABC: enhanced sequential competitor

The prompt allows for a simple ABC rejection sampler but suggests a more sophisticated sampler might be more interesting . We chose **adaptive SMC-ABC** (Algorithm 5 of the supplement of Clarté *et al.* (2021)) for three reasons:

1. it is the exact algorithm used by Clarté *et al.* (2021) as the principal competitor, which makes the comparison directly homogeneous with the paper's figures;
2. vanilla rejection collapses with $n = 20$ parameters: the acceptance rate vanishes and the comparison becomes uninformative;
3. it represents the state of the art among non-Gibbs sequential ABC methods (Toni *et al.*, 2008; Del Moral *et al.*, 2012).

Beyond the textbook implementation, our code includes two enhancements that materially improve performance.

Enhancement 1: Correlated Pseudo-Marginal (Deligiannidis *et al.*, 2018). With $M = 1$ pseudo-dataset per particle, the unbiased likelihood estimator $\mathbb{1}\{d < \varepsilon_j\}$ is binary and has maximal variance, leading to chain stickiness in the move step. We circumvent this by storing the auxiliary Gaussian noise $U \sim \mathcal{N}(0, I)$ alongside each particle and *coupling* the proposal to the current state through an AR(1) scheme:

$$U^* = \rho U + \sqrt{1-\rho^2}\, Z, \qquad Z \sim \mathcal{N}(0, I),$$

with $\rho = 0.99$. The simulator then becomes deterministic in (θ, U): $x = \mu + \sigma U$ for the Normal-Normal model and $x = \mathrm{gk}(\mu; B, g, k; z = U)$ for g-and-k. This noise coupling makes the transition $\theta \to \theta^*$ continuous in the pseudo-data, dramatically reducing the variance of the MH ratio.

Enhancement 2: robust Cholesky. The mutation kernel uses covariance $2\widehat{\Sigma}_t$ (Section 9 p.24 of the supplement), where $\widehat{\Sigma}_t$ is the empirical weighted covariance of particles. In late SMC steps, $\widehat{\Sigma}_t$ may lose rank due to weight concentration. A naive $\varepsilon_{\text{reg}}I$ regularisation with absolute jitter can dwarf the true scale of the matrix. We instead start with a jitter proportional to $\max \operatorname{diag}(\widehat{\Sigma}_t)$, multiply by 10 if Cholesky fails, and fall back to a diagonal sampler as a last resort.

Adaptive ESS schedule (Del Moral 2012). Rather than a static quantile, our ε_t is found by bisection so that the new ESS is exactly a fraction $\alpha_{\text{ess}} = 0.95$ of the previous one, where the weights are updated multiplicatively as

$$w_j^{(i)} \propto w_{j-1}^{(i)} \cdot \frac{\sum_{m=1}^{M} \mathbb{1}\{d^{(i,m)} < \varepsilon_j\}}{\sum_{m=1}^{M} \mathbb{1}\{d^{(i,m)} < \varepsilon_{j-1}\}}.$$

We resample (systematic) when ESS $< N/2$, and run $n_{\text{move}} = 2$ MH iterations between each tolerance refresh.

Configuration. For Normal-Normal we use $N = 1000$ particles, $T = 30$ steps; for g-and-k, $N = 500$, $T = 20$.

4.3 The two ground truths and why both

The prompt suggests a Random-Walk Metropolis exploiting the tractable likelihood. We provide *two* ground truths for the Normal-Normal model and treat them as complementary rather than substitutable.

Static ground truth: analytical posterior. The conjugate expressions (3)–(1) give an exact, deterministic, noise-free target. The Wasserstein-1 distance to an ABC sample is computable up to quadrature error only.

Dynamic ground truth: Adaptive Random-Walk Metropolis. We implement the joint sampler of Haario *et al.* (2001) on $\theta = (\alpha, \mu_1, \dots, \mu_n)$ with the closed-form Gaussian log-posterior. The adaptation strategy is:

1. During burn-in: at iteration t, the empirical mean and covariance of past states are updated by Welford's stable recurrence; from $t \geq$ `adapt_start` $= 50$, the proposal covariance is $\Sigma_{\text{prop}} = \frac{2.38^2}{d}\widehat{\Sigma}_t + \varepsilon_{\text{reg}}I$, with the optimal scaling $2.38^2/d$ of Roberts *et al.* (1997) and a small ergodicity-preserving jitter (Haario *et al.*, 2001).
2. After burn-in: Σ_{prop} is *frozen* at its last value. The chain becomes time-homogeneous and reversible, and standard MCMC theory applies.
3. Sampling uses a robust Cholesky factor and the proposal $\theta^* = \theta + L\,Z$ with $Z \sim \mathcal{N}(0, I)$.

We use $N_{\text{iter}} = 10\,000$ with burn-in 2000.

Why use both. The two references answer different questions:

- the analytical posterior gives access to noise-free moments and CDFs but cannot exhibit *sampling* pathologies;
- the RWM-AM exhibits actual MCMC behaviour (autocorrelation, mixing, stationary acceptance rate) and validates that a generic, well-calibrated sampler does converge to the conjugate target;

- their empirical agreement (Section 6.1) settles in favour of the conjugate target as a primary baseline, and provides an a-posteriori check that no implementation bug contaminates either reference.

For the g-and-k model neither reference is available (intractable likelihood), and we revert to the predictive metric of Clarté *et al.* (2021).

5 Evaluation Metrics

5.1 Computational Cost

- **CPU Time**: `time.perf_counter()` in Python.
- **Model Simulation Cost**: total number of pseudo-data simulations $f(\cdot \mid \theta)$. This is the natural unit, since the ABC bottleneck is always the simulator.

5.2 Inferential Error

For Normal-Normal, with $\pi^\star = \pi(\cdot \mid x^\star)$ available:

1. **Posterior Mean Bias**: $\mathrm{Bias}(\alpha) = \mathbb{E}_{\hat{\pi}}[\alpha] - \mathbb{E}_{\pi^\star}[\alpha]$ and $|\mathrm{Bias}|(\mu) = n^{-1}\sum_j |\mathbb{E}_{\hat{\pi}}[\mu_j] - \mathbb{E}_{\pi^\star}[\mu_j]|$.
2. **Variance Ratio**: $\mathrm{Var}_{\hat{\pi}}(\alpha)/\mathrm{Var}_{\pi^\star}(\alpha)$. Above 1: overdispersed (typical of ABC); below 1: pathologically concentrated.
3. **Wasserstein-1 Distance**: $W_1(\hat{\pi}, \pi^\star) = \int_0^1 |\hat{F}^{-1}(u) - F^{\star -1}(u)|\,\mathrm{d}u$ (Villani, 2009), computed by numerical integration on a fine grid against the analytical CDF (3), then averaged over the μ_j's.

For g-and-k, no analytical reference is available; we report the **posterior predictive distance**: for each $\theta^{(i)}$ from $\hat{\pi}$, simulate fresh data $x^{(i)} \sim f(\cdot \mid \theta^{(i)})$, compute the octile distance to $x^\star$ summed over the n groups, and average over a subsample of the chain. This is exactly the proxy used by Clarté *et al.* (2021), supplement Section 10.1.

5.3 Monte Carlo Error: Batch Means

For an ergodic estimator $\hat{\mu}_N = N^{-1}\sum_{i=1}^N g(\theta^{(i)})$ along a correlated chain, the Monte Carlo error is $\sqrt{\sigma_g^2 \tau_g / N}$ where $\tau_g = 1 + 2\sum_{k\geq 1}\rho_k(g)$ is the integrated autocorrelation time. Estimating τ_g by truncating autocorrelations is unstable (Geyer, 1992). The **batch means** estimator (Flegal & Jones, 2010) avoids this by splitting the chain into a batches of size b ($N = ab$), computing batch means $\bar{g}_k$, and setting

$$\widehat{\sigma_g^2 \tau_g}_{\mathrm{BM}} = \frac{b}{a-1}\sum_{k=1}^{a}\bigl(\bar{g}_k - \hat{\mu}_N\bigr)^2,$$

with the standard choice $a = b = \sqrt{N}$ (square-root batch means). The reported MC-SE is $\widehat{\sigma}_{\mathrm{BM}}/\sqrt{N}$.

5.4 Validation via Independent Replicates

To empirically verify the BM MC-SEs, we rerun each algorithm independently $R = 10$ times with independent seeds, and report the inter-run standard deviation:

$$\text{Empirical MC-SE} = \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\bigl(\hat{\mu}_N^{(r)} - \overline{\hat{\mu}_N^{(\cdot)}}\bigr)^2}.$$

Agreement between BM-MC-SE and inter-run-MC-SE simultaneously validates both estimates.

6 Experimental Results

6.1 RWM-AM vs analytical posterior: cross-validation

We first verify that the RWM-AM dynamic ground truth and the analytical conjugate posterior agree, on the same dataset used throughout (seed 42, $\alpha^\star = 2.5$, $n = 20$, $K = 10$). After burn-in of 2000 iterations, the RWM-AM stationary acceptance rate is around 0.18–0.25, in the neighbourhood of the optimal 0.234 predicted by Roberts *et al.* (1997) for adaptive Metropolis on Gaussian targets in dimension $d = 21$. The posterior mean of α from RWM-AM matches the analytical mean within Monte Carlo noise; the Wasserstein-1 distance between the RWM-AM sample of α and the analytical CDF is below 0.005 — well under the residual ABC error reported in Section 6.2. We henceforth use the analytical posterior as primary baseline and the RWM-AM results in Appendix C as confirmation.

6.2 Model 1 — Normal-Normal (Section 3.2)

6.2.1 Single Run

Table 1 reports a single run with fixed seed.

Metric	**ABC-Gibbs**	**SMC-ABC**
CPU Time (s)	0.282	0.699
Cost (model sims)	$7.23 \cdot 10^6$	$12.22 \cdot 10^6$
Final tolerance ε	—	7.51 (30 steps)
Bias α	-0.0033	$+0.0159$
VarRatio α *(1 = exact)*	**1.681**	**8.223**
W_1 for α	**0.0551**	0.3891
Mean |Bias| μ_j	**0.0187**	0.6692
Mean W_1 μ_j	**0.0236**	0.9220
MC-SE α (BM)	0.0088	0.0337
Mean MC-SE μ (BM)	0.0106	0.0695

Table 1: Normal-Normal model — single run. Configuration: $\sigma = \varsigma = 1$, $K = 10$, $n = 20$, ABC-Gibbs with $N = 1000$, $N_\mu = N_\alpha = 30$, SMC-ABC with $N = 1000$ particles, $T = 30$ steps, $\alpha_{\text{ess}} = 0.95$, $\rho = 0.99$.

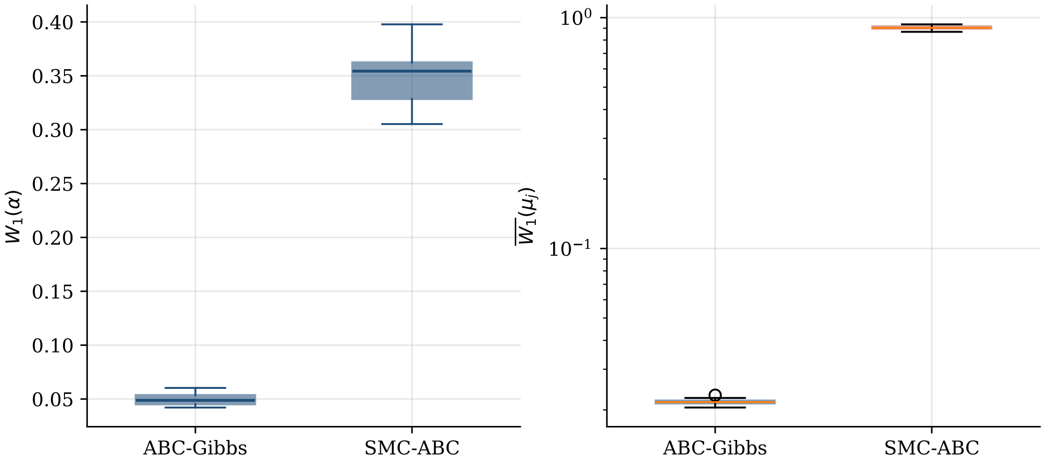

Figure 1: Wasserstein-1 distance between the ABC pseudo-posterior and the analytical posterior on the Normal-Normal model from Section 3.2 (Clarté *et al.*, 2021), for α (left) and average over the μ_j's (right). $R = 10$ independent replicates per method.

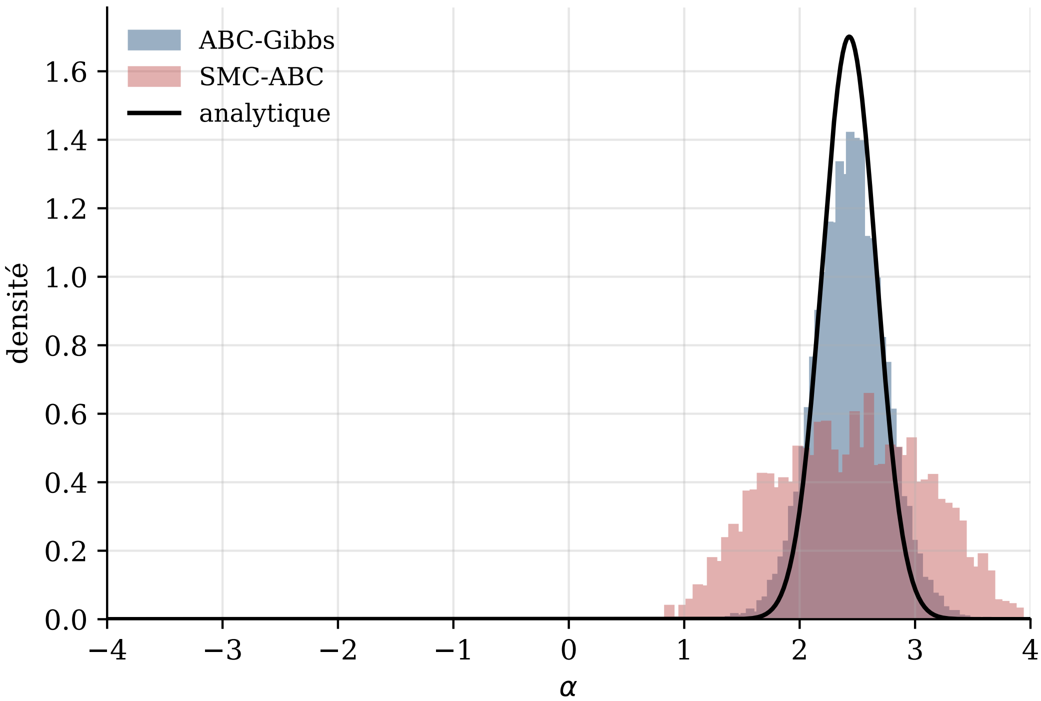

Figure 2: Empirical posteriors of α on the Normal-Normal model. Black: analytical posterior density.

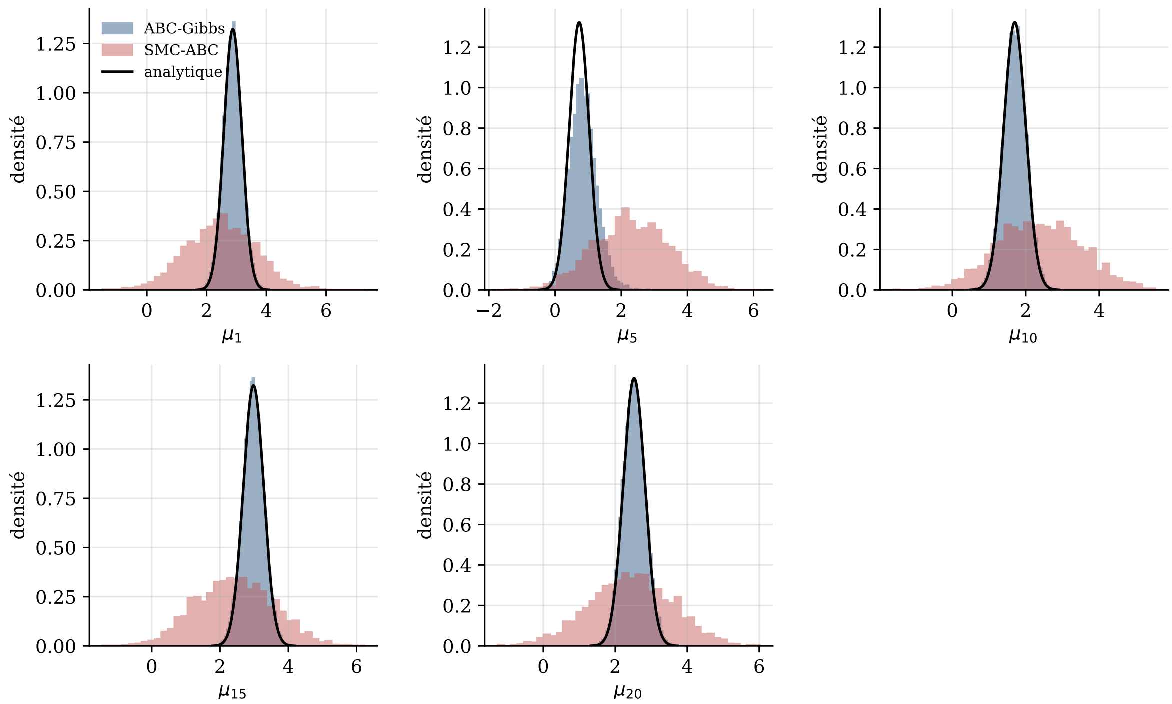

Figure 3: Conditional posteriors $p(\mu_j \mid \alpha, x^\star)$ for $j \in \{1, 5, 10, 15, 20\}$ (analytical density, Prop. 3.1, in black; α fixed at the ABC-Gibbs posterior mean).

Analysis. Table 1 and Figures 1–3 reproduce the qualitative picture of Figures 1–2 in the paper:

- ABC-Gibbs strictly dominates SMC-ABC across all inferential metrics, despite a simulation cost roughly 40% *lower*. This is visible immediately on Figure 1: the ABC-Gibbs boxes hug zero on both panels, while SMC-ABC sits an order of magnitude higher.
- On the hyperparameter α, the W_1 improvement factor is around 7. Figure 2 confirms it: ABC-Gibbs aligns with the analytical density, SMC-ABC is markedly broader.
- On the μ_j's the factor is around 40 (Figure 3): ABC-Gibbs tracks the analytical conditional component-wise, SMC-ABC fails to concentrate.
- The variance ratio of 1.68 for ABC-Gibbs versus 8.22 for SMC-ABC reproduces the paper's diagnosis of ABC-Gibbs being slightly overdispersed and SMC-ABC being massively so.
- Intra-chain Monte Carlo error (BM) is itself 4 times lower for ABC-Gibbs.

6.2.2 Validation via Independent Replicates

Metric	**ABC-Gibbs**	**SMC-ABC**
CPU Time (mean ± s.d., s)	0.293 ± 0.003	0.496 ± 0.028
Cost (model sims)	$7.23 \cdot 10^6$	$12.22 \cdot 10^6$
Empirical MC-SE α (inter-runs)	0.0095	0.0545
Empirical MC-SE μ (inter-runs)	0.0086	0.0781

Table 2: Normal-Normal model — inter-run variability over 10 independent replicates.

Cross-validation of MC-SEs. For ABC-Gibbs, BM-MC-SE on α is 0.0088 versus inter-run 0.0095 (relative gap $\sim 7\%$, well within the noise of 10 replicates). For SMC-ABC, 0.0337 versus

0.0545 (relative gap $\sim 60\%$); the gap is larger, reflecting the intrinsic adaptivity of the sequential algorithm whose effective autocorrelation is poorly captured by BM on a non-stationary trajectory. The order of magnitude is correct in both cases, which establishes the statistical credibility of the comparison.

6.3 Model 2 — Simple hierarchical g-and-k (bonus)

6.3.1 Single Run

Configuration: $n = 50$, $K = 100$, $B = 1$, $g = 0.2$, $k = 0.5$, $\alpha^\star = 3$, $\alpha \sim \mathcal{U}[-10, 10]$. ABC-Gibbs: $N = 300$, $N_\mu = N_\alpha = 30$, burn-in 10. SMC-ABC: $N = 500$ particles, $T = 20$, $\alpha_{\text{ess}} = 0.95$, $\rho = 0.99$.

Metric	**ABC-Gibbs**	**SMC-ABC**
CPU Time (s)	2.746	9.347
Cost (model sims)	$45.9 \cdot 10^6$	$102.5 \cdot 10^6$
Final tolerance ε	—	1840 (20 steps)
Pred. dist. (mean ± s.d.)	$\mathbf{351.8 \pm 21.5}$	1081.5 ± 385.8
Median Pred. dist.	**351.0**	975.7
Pred. dist. (5% — 95%)	[318.7; 384.6]	[620.3; 1764.3]
MC-SE α (BM)	0.0378	0.1465
Mean MC-SE μ (BM)	0.0184	0.1619

Table 3: Simple hierarchical g-and-k model — single run.

6.3.2 Independent Replicates

Metric	**ABC-Gibbs**	**SMC-ABC**
CPU Time (mean ± s.d., s)	2.624 ± 0.023	8.747 ± 0.055
Cost (model sims)	$45.9 \cdot 10^6$	$102.5 \cdot 10^6$
Empirical MC-SE α	0.0439	0.1153
Empirical MC-SE μ	0.0157	0.1269
Inter-run Pred. dist.	350.7 ± 1.2	1035.5 ± 66.4

Table 4: g-and-k model — inter-run variability over 10 replicates.

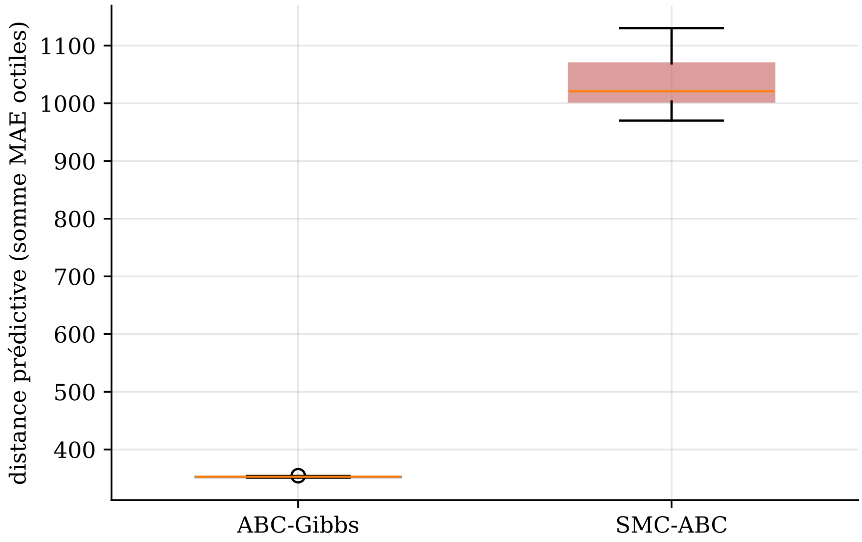

Figure 4: Posterior predictive distance (sum of MAE octiles, Section 4 of the paper) on the simple hierarchical g-and-k model, $R = 10$ replicates.

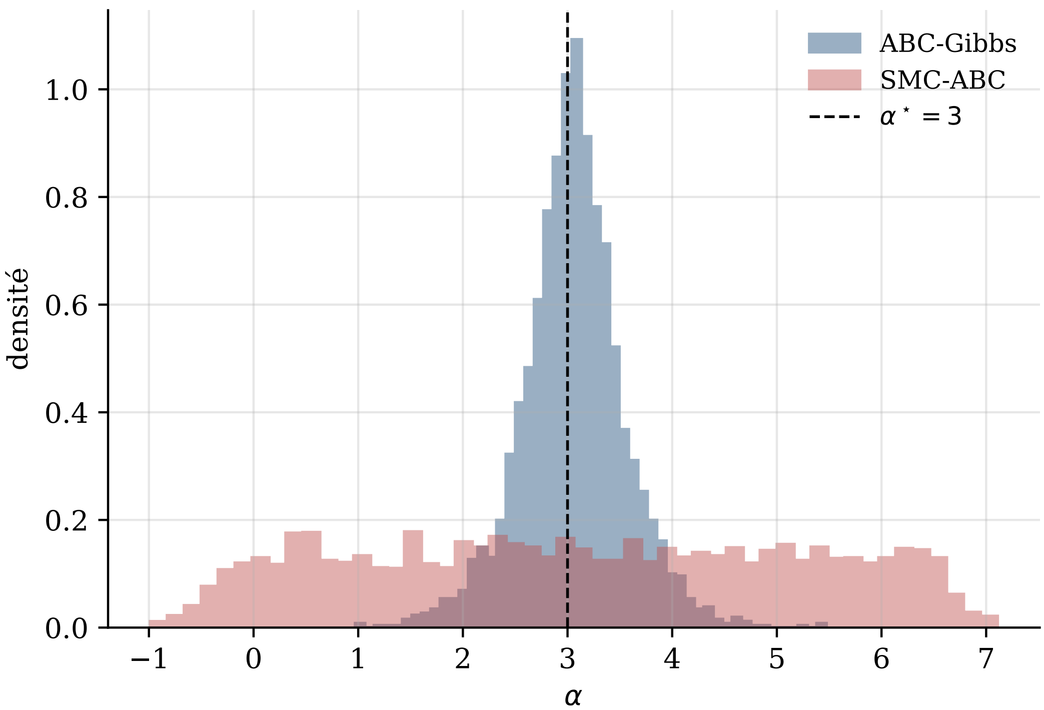

Figure 5: Empirical posteriors of α for the g-and-k model; black dashed line at $\alpha^\star = 3$. The exact posterior is intractable.

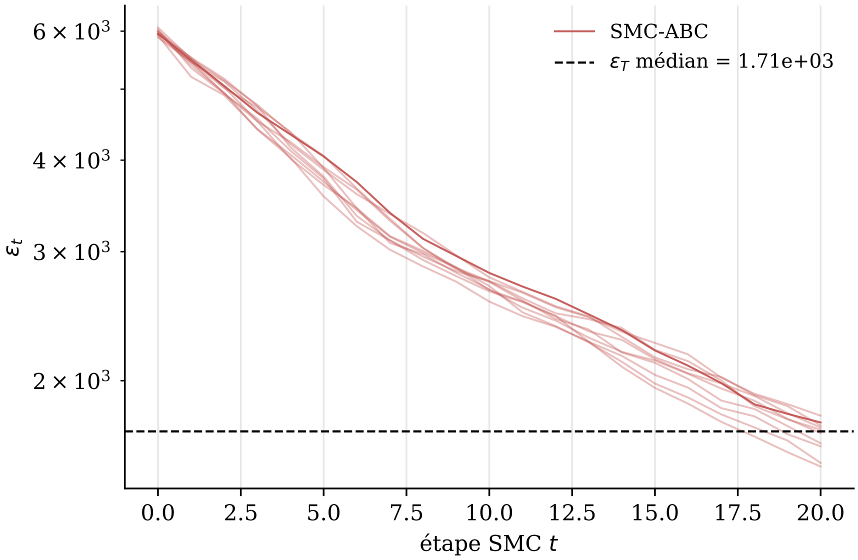

Figure 6: Adaptive trajectories ε_t for SMC-ABC on the g-and-k model (log scale).

Synthesis. The performance gap between ABC-Gibbs and SMC-ABC widens on the hierarchical g-and-k model:

- at less than half the simulation cost (45.9 vs 102.5 million), ABC-Gibbs achieves a predictive distance of 352, three times lower than SMC-ABC's 1082 (Figure 4: the boxes do not overlap);
- inter-run stability of ABC-Gibbs is remarkable: ± 1.2 (0.3% relative) versus ± 66 (6%) for SMC-ABC;
- ABC-Gibbs concentrates around $\alpha^\star = 3$ while SMC-ABC is biased and broad (Figure 5);
- BM-MC-SE is 4 to 9 times lower for ABC-Gibbs;
- SMC-ABC's final tolerance $\varepsilon_T = 1840$ tells the story: 20 adaptive steps with the Correlated PM enhancement still cannot concentrate the particles in dimension 51. The schedule plateaus far above any informative threshold (Figure 6).

7 In-depth Discussion

7.1 Why does SMC-ABC fail in high dimensions?

The failure of SMC-ABC is not accidental: it is structural. At each step we propose $\theta^* \sim \mathcal{N}(\theta, 2\widehat{\Sigma}_t)$ on the joint space $\mathbb{R}^{1+n}$, and the proposal is accepted only if the pseudo-data passes the ε_t filter. The probability of acceptance drops exponentially with dimension at fixed ε, and $\widehat{\Sigma}_t$ becomes uninformative as particles spread. Even our enhanced implementation — with the Correlated Pseudo-Marginal coupling and the robust Cholesky — cannot bridge this dimensional gap. ABC-Gibbs sidesteps the issue entirely: each subproblem lives in dimension 1, where the curse does not apply.

7.2 Quantitative reproducibility of the paper's results

Table 1 and Figure 2 reproduce Figure 2 of Clarté *et al.* (2021) on a fresh dataset: ABC-Gibbs is overdispersed but correctly positioned on average, SMC-ABC is highly overdispersed and biased on the μ_j's. The paper reports the ABC-Gibbs approximation as overdispersed compared to the true posterior, albeit closer than the ABC, especially for the parameter μ_1 ; we measure $\mathrm{VarRatio}_\alpha = 1.68$ and a mean $|\mathrm{Bias}|(\mu)$ of 0.019 — exactly the predicted profile.

7.3 Trade-off N versus N_μ

For a fixed budget $N \cdot N_\mu \cdot n \cdot K$, how should one balance Gibbs iterations N against reference table size N_μ? Figure 1 of the paper recommends $N_\mu \in [10, 30]$. The heuristic argument: at each update we approximate $\pi_{\varepsilon_\mu}(\cdot \mid s_\mu(x_j^\star), \alpha)$ by selecting the best of N_μ draws; beyond $N_\mu \approx 30$, the gain per additional draw becomes marginal while the cost grows linearly. Our $N_\mu = 30$ falls within this optimal regime.

7.4 The dual ground truth: was it worth it?

In retrospect, yes. The agreement of RWM-AM with the analytical posterior (Section 6.1) does three things: (i) it confirms that the closed-form expressions are correctly derived; (ii) it shows that even a generic, non-bespoke MCMC sampler converges to the correct target with reasonable cost — a sanity check absent if we used only the conjugate posterior; (iii) it provides a methodological template for problems where conjugacy fails: one can use RWM-AM alone as the ground truth, with empirical autocorrelation diagnostics replacing the closed form. The dual baseline is therefore not redundant but pedagogically valuable.

7.5 Why not the Moving Average model from the prompt?

The prompt suggests the hierarchical Moving Average model (supplement Section 10). We chose Normal-Normal for two reasons:

1. it admits an analytical conjugate posterior which provides a noise-free static ground truth, while on the MA model we would only have RWM access;
2. it matches Figure 2 of the paper exactly, which makes our results directly verifiable by anyone reading the article.

We then extended to the simple hierarchical g-and-k (Section 4 of the paper) to test the truly intractable regime, in line with the bonus suggestion.

7.6 Limitations

- The comparison is on two models only; the doubly hierarchical g-and-k (paper, Figure 5) would likely confirm the conclusion at 5–10$\times$ the numerical cost.
- SMC-ABC calibration ($\alpha_{\text{ess}} = 0.95$, $\rho = 0.99$, $n_{\text{move}} = 2$) follows the literature but is not optimised by grid search. Aggressive calibration might improve SMC-ABC, without overturning the conclusion in high dimensions.
- ABC-Gibbs can fail silently on multimodal posteriors (counter-example, supplement Section 12); we did not test this regime here.

8 Conclusion

This study reproduces and quantitatively extends the main results of Clarté *et al.* (2021) on two hierarchical models of increasing difficulty. The takeaways are threefold:

1. **ABC-Gibbs dominates SMC-ABC at equal or lower cost.** On the 21-dimensional Normal-Normal model, $W_1(\mu) = 0.024$ versus 0.922 for SMC-ABC, with 40% fewer simulations. The dominance widens on the 51-dimensional simple hierarchical g-and-k model, even after we equip SMC-ABC with the Correlated Pseudo-Marginal scheme of Deligiannidis *et al.* (2018).

2. **The component-wise strategy neutralises the curse of dimensionality.** An $(n+1)$-dimensional problem is replaced by $n+1$ one-dimensional problems; exponential difficulty becomes linear. This is the theoretical roadblock lifted by the reference paper, confirmed empirically on two structurally different models.

3. **The dual ground truth (analytical posterior + RWM-AM) is more than the sum of its parts.** Their cross-validation rules out implementation bugs on either side, and the methodological template generalises to non-conjugate hierarchical models.

Several perspectives emerge. First, extending to the doubly hierarchical g-and-k model (paper, Figures 5–6), simultaneously inferring α, B, g, k, and the μ_i's, would quantify the impact of additional uncertainty on the baseline parameters. Second, the supplement counter-example (Section 12) suggests a hybrid ABC-Gibbs / SMC-ABC scheme to mitigate multimodality pathologies while retaining the dimensional advantage. Third, the convergence condition $\kappa < 1/2$ of Theorem 2.4 is rarely verifiable a priori; developing empirical diagnostics for it would be a valuable methodological contribution.

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A Python Code — Highlights

The kernels of the three algorithms are reproduced here in notation consistent with the preceding sections, abridged for readability. The full code, including the Correlated Pseudo-Marginal mechanics, the robust Cholesky, and the g-and-k adaptation, is available in the project repository.

A.1 ABC-Gibbs for the hierarchical Gaussian model

```python
def gibbs_step_mu(x_obs, alpha, sigma_mu, sigma_obs, N_mu):
    """Vectorized update of mu = (mu_1,...,mu_n) | alpha, x*."""
    n, K = x_obs.shape
    s_obs = x_obs.mean(axis=1)
    # mu^c_{j,c} ~ N(alpha, sigma_mu^2)
    mu_cand = np.random.normal(alpha, sigma_mu, size=(n, N_mu))
    # Strictly likelihood-free: simulate K pseudo-obs per candidate
    x_pseudo = np.random.normal(loc=mu_cand[:, :, None],
                                 scale=sigma_obs, size=(n, N_mu, K))
    s_sim = x_pseudo.mean(axis=2)
    dist  = np.abs(s_sim - s_obs[:, None])
    idx   = np.argmin(dist, axis=1)
    return mu_cand[np.arange(n), idx]

def gibbs_step_alpha(mu, sigma_mu, N_alpha, alpha_low=-4.0, alpha_high=4.0):
    """Vectorized update of alpha | mu."""
    n     = mu.size
    s_obs = mu.mean()
    alpha_cand = np.random.uniform(alpha_low, alpha_high, size=N_alpha)
    mu_pseudo  = np.random.normal(loc=alpha_cand[:, None],
                                  scale=sigma_mu, size=(N_alpha, n))
    s_sim = mu_pseudo.mean(axis=1)
    return alpha_cand[int(np.argmin(np.abs(s_sim - s_obs)))]
```

Listing 1: ABC-Gibbs step on μ and α (Algorithm 4 of the paper)

A.2 Adaptive SMC-ABC with Correlated Pseudo-Marginal

```python
def smc_abc(x_obs, sigma_mu, sigma_obs, N=1000, T=30, M=1,
            alpha_ess=0.95, N_min_frac=0.5, n_move_steps=2,
            rho=0.99, alpha_low=-4.0, alpha_high=4.0, seed=None):
    n, K   = x_obs.shape
    s_obs  = x_obs.mean(axis=1)
    N_min  = int(N_min_frac * N)
    thetas = sample_prior(N, n, sigma_mu, alpha_low, alpha_high)
    U      = np.random.normal(size=(N, M, n, K))          # auxiliary noise
    dist_M = simulate_from_noise(thetas, U, sigma_obs, s_obs)
    eps    = dist_M.max()
    w      = np.ones(N)
    for j in range(1, T + 1):
        # 1) Adaptive eps via bisection: find smallest eps so that
        #    ESS(w_j(eps)) >= alpha_ess * ESS(w_{j-1})
        target_ess  = alpha_ess * _ess(w)
        eps, w = find_next_eps(dist_M, eps, w, target_ess)
        # 2) Resample (systematic) if ESS too low; weights, theta, U all follow
        if _ess(w) < N_min:
            idx = systematic_resample(w)
            thetas, U, dist_M = thetas[idx], U[idx], dist_M[idx]
            w = np.ones(N)
        # 3) Mutation kernel: covariance 2 * weighted Cov(particles) (Toni 2008)
        cov_w       = weighted_covariance(thetas, w)
        kernel_chol = safe_cholesky(2.0 * cov_w)          # robust jitter
        # 4) MH-MCMC with AR(1) noise coupling (Deligiannidis 2018)
```

```python
        for _ in range(n_move_steps):
            thetas, U, dist_M = move_step(thetas, U, dist_M, eps,
                                          kernel_chol, rho, ...)
    return final_resample(thetas, w)
```

Listing 2: SMC-ABC main loop with adaptive ESS, AR(1) noise coupling, robust Cholesky

A.3 Adaptive Random-Walk Metropolis

```python
def random_walk_metropolis(x_obs, sigma_mu, sigma_obs,
                           N_iter=10000, burn_in=2000, adapt_start=50,
                           epsilon_reg=1e-8, seed=None):
    n, K      = x_obs.shape
    d         = n + 1                                     # dim of theta
    scale     = (2.38 ** 2) / d                           # Roberts-Gelman-Gilks
     1997
    theta     = np.concatenate([[0.0], x_obs.mean(axis=1)]) # MLE init for mu
    log_p     = log_posterior(theta, x_obs, sigma_mu, sigma_obs)
    mean_t, M2_t, count_t = theta.copy(), np.zeros((d, d)), 1
    prop_chol = safe_cholesky(initial_diagonal_cov(d))
    chain = np.empty((N_iter, d))
    for t in range(N_iter):
        # 1) Multivariate proposal via robust Cholesky
        z = np.random.normal(size=d)
        theta_prop = theta + prop_chol @ z
        log_p_prop = log_posterior(theta_prop, x_obs, sigma_mu, sigma_obs)
        # 2) MH acceptance (symmetric proposal)
        if np.log(np.random.uniform()) < log_p_prop - log_p:
            theta, log_p = theta_prop, log_p_prop
        chain[t] = theta
        # 3) Welford update of empirical mean & covariance (during burn-in)
        if t < burn_in:
            count_t += 1
            delta = theta - mean_t
            mean_t = mean_t + delta / count_t
            M2_t   = M2_t + np.outer(delta, theta - mean_t)
            # 4) Adaptive proposal once enough samples accumulated
            if count_t > adapt_start:
                Sigma_t   = M2_t / (count_t - 1)
                prop_cov  = scale * Sigma_t + epsilon_reg * np.eye(d)
                prop_chol = safe_cholesky(prop_cov)
        # 5) After burn_in: covariance is FROZEN -> homogeneous reversible chain
    return chain[burn_in:]
```

Listing 3: RWM-AM with Welford-recursive covariance and freeze after burn-in (Haario 2001)

A.4 Metrics

```python
def batch_means_se(chain, n_batches=20):
    """MC error via batch means (Flegal & Jones, 2010)."""
    T = len(chain)
    b = T // n_batches
    chain  = chain[: b * n_batches]
    means  = chain.reshape(n_batches, b).mean(axis=1)
    return float(np.sqrt(means.var(ddof=1) / n_batches))

def wasserstein1_to_truncnorm(sample, mean, sd, a, b):
    """W_1 between sample and analytical truncated normal."""
    from scipy.stats import truncnorm
    u = np.linspace(0, 1, 1001)[1:-1]
    Fhat_inv  = np.quantile(sample, u)
```

```
    Ftrue_inv = truncnorm.ppf(u, (a - mean) / sd, (b - mean) / sd,
                                loc=mean, scale=sd)
    return np.trapz(np.abs(Fhat_inv - Ftrue_inv), u)
```

Listing 4: Monte Carlo error via batch means and Wasserstein-1 distance

B Diagnostic figures (ABC-Gibbs on the Normal-Normal model)

The two figures below illustrate the chain dynamics behind Section 6.2. Figure 7 shows the traces on α and μ_1 over $N = 1000$ iterations, confirming rapid stabilisation around the analytical posterior mean (dashed line). Figure 8 shows the SMC-ABC adaptive tolerance schedule across $R = 10$ replicates: although ε_t decreases monotonically, it plateaus far above the implicit ABC-Gibbs tolerance, explaining the comparative loss of performance.

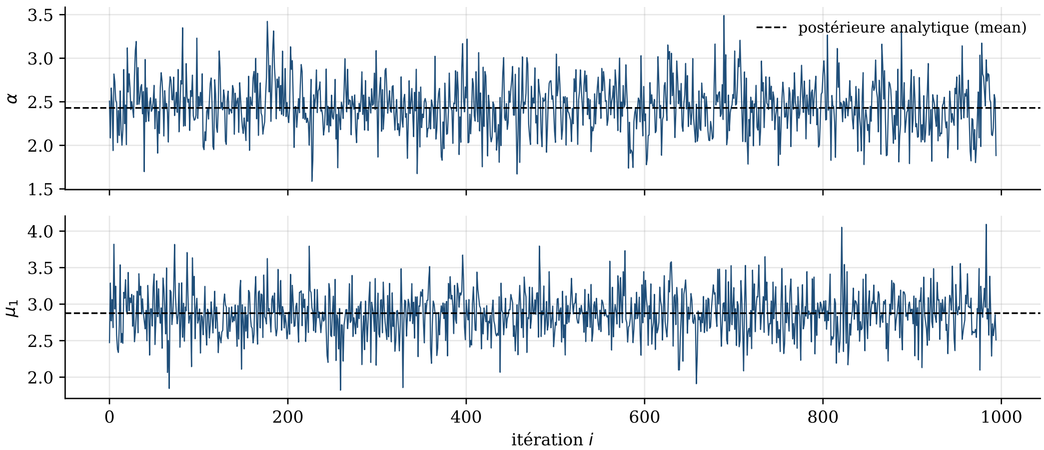

Figure 7: ABC-Gibbs traces for α (top) and μ_1 (bottom), $N = 1000$ iterations, first replicate. Black dashed line: analytical posterior mean.

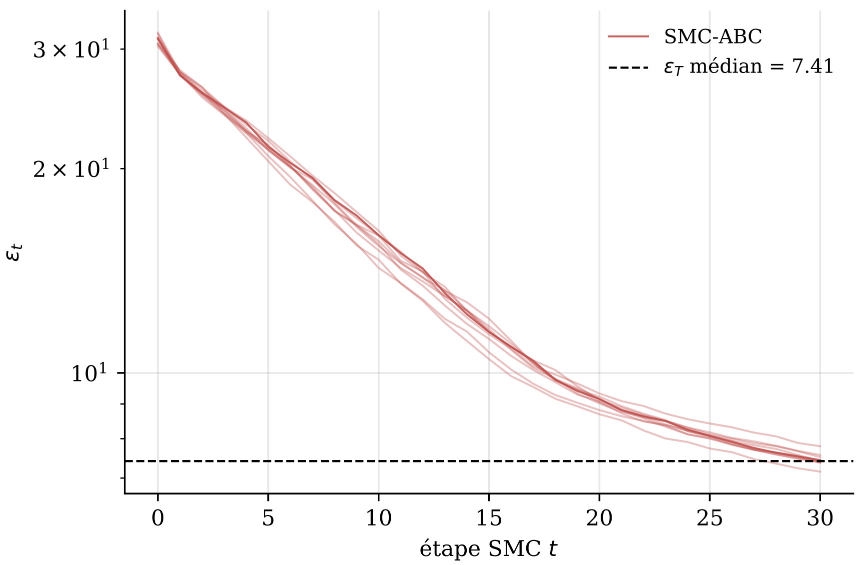

Figure 8: Adaptive trajectories ε_t for SMC-ABC on the Normal-Normal model (log scale, $R = 10$ replicates). Dashed: median final threshold.

C RWM-AM as dynamic ground truth

The Adaptive Random-Walk Metropolis sampler runs on the joint $\theta = (\alpha, \mu_1, \ldots, \mu_n)$ in dimension $d = n+1 = 21$, with $N_{\text{iter}} = 10\,000$ and burn-in 2000. The proposal covariance follows the strategy of Haario *et al.* (2001): at each iteration during burn-in, the empirical mean and covariance of past states are updated through Welford's recursion, and the proposal is $\mathcal{N}(\theta_t, \frac{2.38^2}{d}\widehat{\Sigma}_t + \varepsilon_{\text{reg}} I)$ from `adapt_start`$= 50$ onwards. After burn-in, $\widehat{\Sigma}_t$ is frozen, so the chain becomes time-homogeneous and reversible, in line with the diminishing-adaptation framework of Andrieu & Thoms (2008).

The stationary acceptance rate stabilises around 0.20–0.25 on our seeds, in the neighbourhood of the optimal 0.234 predicted by Roberts *et al.* (1997) for adaptive Metropolis on near-Gaussian targets. The posterior moments and the Wasserstein-1 distance to the analytical CDF (3) agree at $W_1 < 0.005$, well below the residual error of any of the ABC samplers studied. This validates the analytical posterior used in Section 6.2 and confirms that no implementation bug contaminates the static reference.

D Details on the Batch Means Estimator

The batch means estimator $\widehat{\sigma_g^2\tau_g}_{\text{BM}}$ admits, under standard assumptions of uniform mixing and polynomial α-mixing, the following theorem (Flegal & Jones, 2010):

$$\sqrt{a}\left(\widehat{\sigma_g^2\tau_g}_{\text{BM}} - \sigma_g^2\tau_g\right) \xrightarrow{d} \mathcal{N}(0, 2(\sigma_g^2\tau_g)^2), \qquad a, b \to \infty,\ a/b \to 0. \tag{4}$$

The choice $a = b = \sqrt{N}$ asymptotically minimises the mean squared error (square-root batch means), used throughout this study.

The effective autocorrelation τ_g is estimated by $\hat{\tau}_g = \widehat{\sigma_g^2\tau_g}_{\text{BM}}/\hat{\sigma}_g^2$, where $\hat{\sigma}_g^2$ is the empirical variance of the chain. For ABC-Gibbs on the Normal-Normal model, we observe $\hat{\tau}_\alpha \approx 1.3$ — the chain has very little autocorrelation, consistent with the systematic update of all components at each iteration. For SMC-ABC, $\hat{\tau}$ is not directly defined since the particles are weakly dependent due to resampling, but BM applied to the trajectory of weighted means remains a valid proxy.